Yanhe Ren

Machine Learning / Research Intern | Summer 2027 (May-Aug)

ryhmtxx.github.io | yanheren@asu.edu | 949-629-5219

EDUCATION

Arizona State University | Ph.D. in Statistics — Aug 2025 - Present | Expected Jun 2029

University of California, Irvine | M.S. in Electrical and Computer Engineering — Sep 2022 - Jun 2024

East China University of Science and Technology | B.Eng. in Automation — Sep 2018 - Jun 2022

RESEARCH EXPERIENCE

Arizona State University | Doctoral Research

FlowPINN: One-Step Generative Modeling

- Proposed FlowPINN, learning neural flow maps through balanced ODE residual minimization at the map's output to generate samples in one network evaluation.
- Achieved **117x faster sampling** than a 100-evaluation flow-matching solver on MNIST using a UNet, with comparable FID (24.29 vs. 24.23) across five seeds.
- Achieved one-step CIFAR-10 **FID of 8.69**, outperforming MeanFlow (9.88) and Shortcut (23.65) with the same backbone and training-step budget.

Uncertainty Quantification for Flow Matching

- Proposed sampler-agnostic numerical-error bars from each sampler's Jacobian, with two scale parameters per sampler and step count calibrated offline against probabilistic-solver endpoints.
- Developed a parallel-in-time probabilistic ODE solver in log-SNR time using a Gauss-Markov prior and Gauss-Newton conditioning across all time nodes.
- Improved CIFAR-10 FID from **11.05 to 5.13** through three rounds of probabilistic refinement and halved MNIST median endpoint error from 7.2 to 3.6.

UC Irvine | Research & Data Analyst, Annie Qu's Lab — Apr 2024 - Apr 2025

- Built and released an iOS/Android EMA app and analytics pipeline for the SMILES study, supporting **110k+ events** using Flutter, Flask, MySQL, Gunicorn, and Nginx.
- Integrated Oura Ring data, implemented SQL constraints and outlier checks, and produced de-identified datasets and cohort dashboards for longitudinal stress, sleep, and activity analysis.
- Prototyped a multi-turn medical RAG assistant using LLaMA, FAISS, and M2-BERT embeddings; built retrieval evaluation with Recall@5/MRR and audited unsupported responses.
- Curated a **1.7M-abstract** arXiv dataset with titles, categories, and timestamps for dynamic topic modeling.

PUBLICATIONS AND MANUSCRIPTS

FlowPINN: Physics-Informed Neural Flow Maps for One-Step Generation.
Yanhe Ren, Zeyuan Yu, Shiwei Lan
Under review at ICLR 2027.

Uncertainty Quantification for Flow Matching via Probabilistic Solver.
Yanhe Ren, Keyan Wu, Shiwei Lan
Manuscript under review.

Context-Sensitive Stress Prediction from Wearable and EMA Data for Just-in-Time Adaptive Interventions.
Cadence Pinkerton, **Yanhe Ren**, Yuqing Guo, Annie Qu
STAI-X 2026, Poster Track.

TECHNICAL SKILLS

Programming: Python, SQL, R

Methods: Generative modeling, uncertainty quantification, Bayesian statistics, retrieval-augmented generation

Tools: PyTorch, Linux, Git, MySQL, Flask